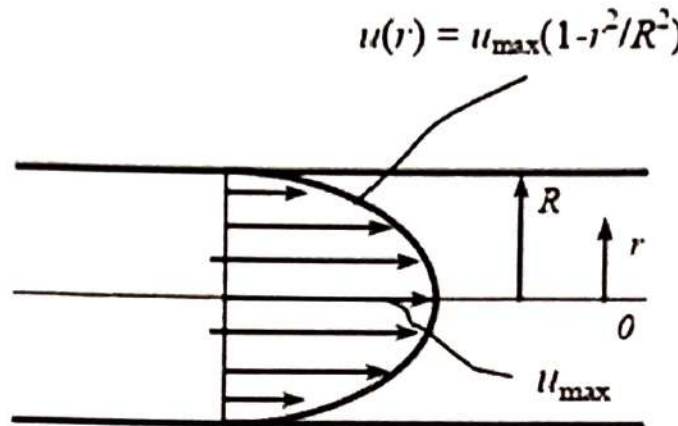


Duration: 40 mins

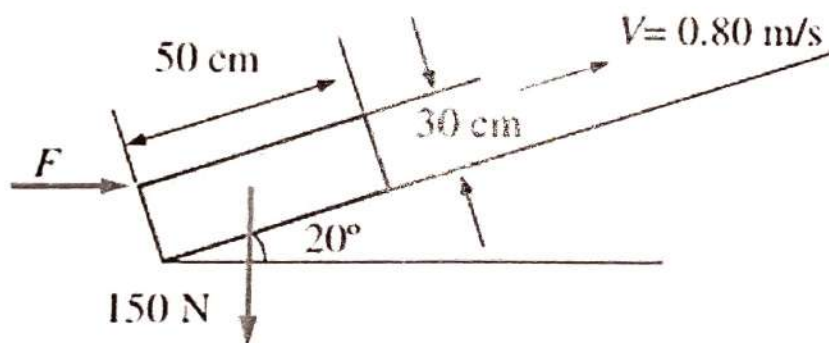
Max Marks: 20

Q1. Water is moving through a pipe with velocity profile shown in figure below. $U_{\max}$ is maximum velocity 3 m/s. Calculate the maximum shear stress and minimum shear stress and corresponding locations if kinematic viscosity of water is $1.0034 \text{ mm}^2/\text{s}$ and density is 0.9982 g/cc . What is the frictional drag force exerted by liquid on the pipe, if same velocity profile exists along full length of the pipe? Radius of the pipe (R) is 0.08 m and Length is 30 m.



[6]

Q2. A 50-cm x 30-cm x 20-cm block weighing 150 N is to be moved at a constant velocity of 0.80 m/s (moved upwards along the slope) on an inclined surface with a friction coefficient of 0.27. (a) Determine the force F that needs to be applied in the horizontal direction. (b) If a 0.40-mm-thick oil film with a dynamic viscosity of 0.012 Pa.s is applied between the block and inclined surface, determine the percent reduction in the required force.



[6]

Q3. Consider a 0.2-mm diameter air bubble in a liquid. Determine the pressure difference between the inside and outside of the air bubble if the surface tension at the air-liquid interface is (a) 0.080 N/m and (b) 0.12 N/m. Hint: Only one surface between air bubble and liquid.

[4]

Q4. Determine the maximum diameter of a steel ball that would float on water at 20°C due to surface tension. What would your answer be for an aluminum ball? Take the densities of steel and aluminum balls to be 7800 kg/m^3 and 2700 kg/m^3 , respectively. Assume contact angle is zero degrees.

[4]